

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Science — Semester I Examination, 2023-24

Paper No. MTB-102: Geometry

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 067936

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) Find the centre and radius of the circle $r = 3(\sqrt{3} \cos \theta + \sin \theta)$.
- (b) Obtain the points on the conic $\frac{15}{r} = 3 + 5 \cos \theta$ whose radius vector is 3.
- (c) Find the distance between two points $(2, \pi/3)$ and $(3, \pi/6)$ in polar coordinates.
- (d) Find the angle between the planes $\mathbf{r} \cdot (3\hat{i} - 4\hat{j} + 5\hat{k}) = 3$ and $\mathbf{r} \cdot (2\hat{i} - \hat{j} - 2\hat{k}) = 2$.
- (e) Find the intercepts made on the axes by the plane $\mathbf{r} \cdot (2\hat{i} + 3\hat{j} - 2\hat{k}) = 6$.
- (f) Write the line $\frac{x-1}{2} = y+1 = \frac{z-3}{4}$ in vector form.
- (g) Find the centre of the circle $x^2 + y^2 + z^2 = 6, z = 2$.
- (h) Define *reciprocal cones*.
- (i) Define *right circular cylinder*.
- (j) Find the equation of the normal to the conicoid $3x^2 - y^2 + 2z^2 = 2$ at the point $(1, -3, 2)$.
- (k) The plane $x + 2y - 2z = 4$ touches the paraboloid $3x^2 + 4y^2 = 24z$. Find the point of contact.

Question 2.

- (a) Define tangent at a point of a conic. Find the equation of the chord joining (r_1, α) and (r_1, β) on the conic $\ell/r = 1 + e \cos \theta$. [6]
- (b) Show that the director circle of the conic $\ell/r = 1 + e \cos \theta$ is $\ell^2(1 - e^2) + 2e\ell r \cos \theta - 2r^2 = 0$. [6]

Question 3.

- (a) A variable plane at constant distance p from the origin meets the axes in A, B, C . Show the locus of the centroid of tetrahedron $OABC$ is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$. [6]
- (b) Find the equation of the line through position vector $3\hat{i} + 2\hat{j} - 4\hat{k}$, parallel to both $\mathbf{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 5$ and $\mathbf{r} \cdot (3\hat{i} + \hat{j} - 2\hat{k}) = 6$. [6]
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Question 4.

- (a) Find the length and equation of the shortest distance between $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + s\mathbf{d}$. [6]
- (b) Find the equation of the sphere through $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$ with smallest radius. [6]
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Question 5.

- (a) Prove that $x + 2y - z = 4$ cuts $x^2 + y^2 + z^2 - x + z - 2 = 0$ in a circle of radius unity. Find the sphere with this circle as a great circle. [6]
- (b) Find the cone with vertex $(-2, 4, 3)$ and guiding curve $3x + 2y = 0$, $y^2 + z^2 = 0$. [6]
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Question 6.

- (a) Find the enveloping cone of $x^2 + y^2 + z^2 - 3x + 2y - 4 = 0$ with vertex $(2, 1, 3)$. [6]
- (b) Find the right circular cylinder of radius 3 with axis $\frac{x-2}{1} = \frac{y-1}{1} = \frac{z+3}{1}$. [6]
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Question 7.

- (a) Find the tangent planes of $2x^2 - 6y^2 + 3z^2 = 5$ passing through the line $x + 9y - 3z = 0$, $3x - 3y + 6z = 5$. [6]
- (b) Prove that enveloping cylinders of $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ whose generators are parallel to $x = 0$, $\frac{y}{b} = \frac{z}{c}$, meet $z = 0$ in circles. [6]
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